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Which directory is that bash script in?

There may be times when you need to know the actual location a BASH script is located within the script. This can be done with a combination of the \$0 value and the dirname command.

The \$0 value

A BASH script makes the first command line argument available as \$1, the second as \$2 and so on. The command run, exactly as it was called but without the command line arguments, is stored in \$0

The dirname command

The dirname command returns the directory name part of a filename. Note that the directory and/or file do not actually exist; dirname simply strips the last component of a file path.

For example, "dirname /a/b/c" will echo "/a/b", "dirname ../a/b/c" will echo "../a/b", "dirname ../" will echo "." and so on.

Putting it all together

The directory the BASH script is located can be retrieved using dirname \$0 like so:

```
DIRECTORY=`dirname $0`
```

BUT note that this may be a relative path and not necessarily an absolute one, depending how the script is called. Take the following script, for example, which saves the directory name to a variable for later use, and then echos it:

```
#!/bin/bash
DIRECTORY=`dirname $0`
echo $DIRECTORY
```

If the file is saved at /home/chris/bin/test1.sh, the permissions are changed so it can be executed (chmod 0700 ~/bin/test1.sh) and it is run from /home/chris like so:

```
/home/chris/bin/test1.sh
```

then it will echo this:

```
/home/chris/bin
```

It it is run like so:

```
bin/test1.sh
```

then it will echo this:

```
bin
```

Because the path used to run the script in the second example is a relative path, "dirname \$0" only echos "bin", which may not be suitable.

Always return the absolute path

To always return the absolute path to the directory the script is located, change to the scripts directory and get the dirname like so, again saving the value to a variable for later use and then echoing it:

```
#!/bin/bash
DIRECTORY=$(cd `dirname $0` && pwd)
echo $DIRECTORY
```

If this script is saved as /home/chris/bin/test2.sh , then whether it is run as /home/chris/bin/test2.sh or bin/test2.sh or by some other path, it will always echo:

```
/home/chris/bin
```

Note that although the change directory command (cd) is used, the script will not change directory and any other calls within it are still relative to the current working directory.

If we were running the above script in /home/chris, then calling "pwd" in the next line would still echo "/home/chris"

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